v1.0
2026

FYP SYSTEM ARCHITECTURE · DESIGN DOCUMENT

Cogdoc

A research-assisted writing IDE that uses **productive AI friction** to mitigate cognitive atrophy in the age of generative AI.

Modular monolith · React + Tiptap · FastAPI · Supabase pgvector · Gemini · SSE.

Seven architectural decisions, three user journeys, sixteen use cases, ten entities at third normal form, and a provider abstraction layer that decouples four AI engines — Interceptor, Socratic Spacer, Citation Sentinel, Living Margin — from any single LLM vendor. The system never writes the user's prose; it interrupts, questions, challenges, scaffolds, and verifies.

7	4	10	60
ARCHITECTURAL DECISIONS	AI ENGINES	DB TABLES AT 3NF	DEV DAYS (RAD)

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